

Course Management System (CMS)

A full-stack web application for managing courses, users, and enrollments. Built with PHP, MySQL, and vanilla JavaScript.

Login Credentials:

Username: np03cs4a240036

Password: jKX72aMMIS

Server Link:

https://student.heraldcollege.edu.np/~np03cs4a240036/Full_stack/public/login.php

Overview

The Course Management System allows **administrators** to create and manage courses and instructors, **instructors** to be assigned to courses, and **students** to browse courses, enroll, and view their enrolled courses. The application includes authentication, role-based access control, CSRF protection, and a live search feature.

Features

- **User roles:** Admin, Student, Instructor
- **Authentication:** Login, registration, logout with session handling and password hashing (bcrypt)
- **Security:** CSRF tokens on forms, prepared statements (SQL injection prevention), output escaping
- **Courses:** List, add, edit, delete (admin only); each course can have an optional instructor
- **Enrollments:** Students can enroll in courses; one enrollment per user per course
- **Live search:** Ajax-based search by course title or instructor name without page reload
- **Responsive layout:** CSS-based styling with a consistent theme

Technology Stack

Layer	Technology
Backend	PHP 7.4+
Database	MySQL 5.7+ / MariaDB
Frontend	HTML5, CSS3, JavaScript
Server	Apache (e.g. XAMPP)

Project Structure

```

Full_stack/
├── config/
│   ├── db.php
│   └── setup_admin.php
├── includes/
│   ├── functions.php
│   ├── header.php
│   └── footer.php
├── assets/
│   ├── css/style.css
│   └── js/search.js
├── public/
│   ├── index.php
│   ├── login.php, register.php, logout.php
│   ├── dashboard.php
│   ├── my_courses.php
│   ├── enroll.php, search.php
│   ├── add_course.php, edit_course.php, delete_course.php
│   └── add_instructor.php
├── database.sql
├── add_enrollments_table.sql
├── add_instructor_support.sql
└── README.md

```

Database Schema

users — id, username (unique), password (bcrypt), role (admin|student|instructor), created_at

courses — id, title, description, category, price, instructor_id (FK → users), created_at

enrollments — id, user_id, course_id, enrolled_at; unique (user_id, course_id)

Installation

1. **Requirements:** PHP 7.4+ with PDO MySQL, MySQL 5.7+ or MariaDB, web server (e.g. Apache/XAMPP).
2. **Database:** Create database, run database.sql, then add_enrollments_table.sql and add_instructor_support.sql if needed.
3. **Config:** Edit config/db.php with host, database name, user, and password.
4. **Run:** Place project under document root and open http://localhost/Full_stack/public/ (or your URL).

Default seed users (change password in production): admin, student, instructor1 — password often password for development.

Usage

Guests: Login, Register. **Students:** Dashboard, search, enroll, My Courses. **Admins:** Add/edit/delete courses, add instructors.

Known Issues:

Currently there is problem of duplicates users. This can be worked upon in further refinement of the project.